

EE 252L Signals & Systems Lab

Spring 2022

Reference Document:

Creating Function in MATLAB

Functions are a special type of m-file that can accept inputs (matrices and vectors) and also return outputs. In the upcoming task we are going to create our own function. Below is an example code given for a custom sine function. A new function is created through an m-file where the first word of first line has to be the keyword function, followed by the details of outputs, function name, and input arguments. On the left side of equal sign in square brackets are the output arguments and on the right sign of the equal is the name of the function, `sin_x`, and the input arguments in round brackets.

Conventionally, the first line is followed by comments describing the purpose of function, its inputs, and outputs. The function body (code) is included between the starting keyword function and the ending keyword end.

- 💡 Once you have written your code, **save the file with the same filename as function name otherwise it won't work.**
- 💡 **Do not run the file you will only save your function**

```
function [y]=x_square(x)
% x_square function to generate a square signal of the input array
% y = x^2
% x=array of input signal
% y = array of square values of the array 'x'

x = -2:1:2;
y=x.^2 %using dot operator for element by element array operation
end
```

Call this function in your program.

- 💡 Never use a variable with the same name as a MATLAB function or command. If you do so, that function or command will become inaccessible. Also, if there is more than one function or command with the same name, the *first* one found on the search path will be executed and all of the others will be inaccessible.
- 💡 Never create an M-file with the same name as a MATLAB function or command.